

3 The matrices **M** and **N** are given by

$$\mathbf{M} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \text{ and } \mathbf{N} = \begin{pmatrix} b & -a \\ a & b \end{pmatrix} \text{ where } a \text{ and } b \text{ are positive constants.}$$

(a) Given that $\mathbf{M}^2 = \mathbf{N}$, determine the exact values of a and b . **[4]**

(b) Hence state the transformations of the plane associated with matrices **M** and **N**. **[3]**